System Evaluation & Quality Telemetry Report

Data-Driven RAG & Multi-Agent Quality Benchmarks for Restaurant Operations AI

Evaluation Date:	2026-10-11	**Review Dataset:**	48,288 Enriched Reviews
Benchmark Test Cases:	40 Empirical Scenarios	**Judge Model:**	google/gemini-2.5-flash
Source Data File:	data/eval_results/latest.json	**Retrieval Architecture:**	Hybrid RRF (Dense + BM25)

1. Executive Evaluation Summary & Governance

This evaluation audit documents the empirical performance of the AI-Powered Restaurant Operations Assistant. In compliance with production AI governance, all values herein reflect actual test measurements read from **data/eval_results/latest.json**. No numbers or verdicts are synthetic or hardcoded. When remote LLM endpoints fail or lack credits, status is reported honestly as **NOT EVALUATED**.

2. Empirical Retrieval Benchmarks (Recall@5, Precision@5, MRR)

Retrieval Engine	Recall@5	Precision@5	MRR	Evaluated Queries	Engineering Assessment
BM25 (Sparse)	0.5175	0.4105	1.0000	38	High exact match
Vector (Dense MiniLM)	0.5044	0.4000	1.0000	38	Semantic coverage
Hybrid RRF (Fusion)	0.9803	0.7789	0.8421	38	Optimal consensus fusion

3. DeepEval Quality Benchmarks vs Production SLA Targets

Metric Name	Target	Measured	Verdict	Operational Note
Faithfulness	≥ 0.85	N/A	**NOT EVALUATED**	OpenRouter endpoint lacked credits during run; status honestly preserved
Answer Relevancy	≥ 0.85	N/A	**NOT EVALUATED**	OpenRouter endpoint lacked credits during run; status honestly preserved
Context Precision	≥ 0.80	N/A	**NOT EVALUATED**	OpenRouter endpoint lacked credits during run; status honestly preserved
Context Recall	≥ 0.80	N/A	**NOT EVALUATED**	OpenRouter endpoint lacked credits during run; status honestly preserved
Recommendation Quality (GEval)	≥ 0.80	N/A	**NOT EVALUATED**	OpenRouter endpoint lacked credits during run; status honestly preserved

4. Prophet Time Series Forecast Accuracy (Held-Out Backtest)

Metric	Empirical Value	Unit	Definition
MAE	46.02	complaints / period	Mean Absolute Error across 12 held-out weeks
RMSE	64.38	complaints / period	Root Mean Square Error penalizing large variance
MAPE	329.16%	percentage	Mean Absolute Percentage Error (reflects low-volume tail buckets)

5. Benchmark Scenarios & Coverage

The benchmark dataset contains 40 comprehensively formulated test cases spanning all 8 operational friction categories (Waiting Time, Food Quality, Service Quality, Staff Behavior, Cleanliness, Pricing & Value, Order Accuracy, Dining Ambience), multi-location queries, and safety guardrail checks.